

From: [Saif, Linda](#)
To: [Liu, Shan-Lu](#); lishan_su@med.unc.edu
Subject: FW: A commentary on 2019 nCoV vs lab engineered viruses
Date: Wednesday, February 12, 2020 1:28:39 PM
Attachments: [EMI-2019-nCoV_Commentary_LJS_SLL_Refs-rsb.docx](#)

Hi

Please note that Ralph made these changes on an earlier copy sent to him so hopefully the 2 of you can incorporate them into the updated draft I sent this AM!

Regards,

Linda

Linda J. Saif, PhD
Distinguished University Professor
Food Animal Health Research Program
OARDC/The Ohio State University
1680 Madison Ave
Wooster, Oh 44691

From: "rbaric@email.unc.edu" <rbaric@email.unc.edu>
Date: Wednesday, February 12, 2020 12:32 PM
To: "Su, Lishan" <lishan_su@med.unc.edu>
Cc: Linda Saif <saif.2@osu.edu>
Subject: RE: A commentary on 2019 nCoV vs lab engineered viruses

My comments. I've included an excel file comparing the differences in the genome length sequences of the parental and chimeric viruses. Also made some text changes. I think the community needs to write these editorials and I thank you for your efforts. ralph

From: Su, Lishan <lishan_su@med.unc.edu>
Sent: Wednesday, February 12, 2020 10:11 AM
To: Baric, Ralph S <rbaric@email.unc.edu>
Subject: Re: A commentary on 2019 nCoV vs lab engineered viruses

Hi Ralph:

We are trying to finish it and had no plan to get you too involved, but I do value your input. It is almost final and we are also getting comments from Perlman and Weiss.

Thanks,

-Lishan

From: "Baric, Ralph S" <rbaric@email.unc.edu>
Date: Wednesday, February 12, 2020 at 10:02 AM
To: "Su, Lishan" <lishan_su@med.unc.edu>

Subject: RE: A commentary on 2019 nCoV vs lab engineered viruses

sure, but don't want to be cited in as having commented prior to submission.

From: Su, Lishan <lishan_su@med.unc.edu>

Sent: Wednesday, February 12, 2020 1:12 AM

To: Baric, Ralph S <rbaric@email.unc.edu>

Subject: A commentary on 2019 nCoV vs lab engineered viruses

Hi Ralph:

In response to the EMI journal editor's request, Drs. Shan-Lu Liu, Lin Saif and myself are writing a commentary (1-2 pages) to dispute the rumors of 2019 nCoV origin. Will you be interested, and have time, to have a quick read/comment? Please let me know if you have time.

Tentative Title: Is 2019-nCoV laboratory origin?

Thanks!

-Lishan

From: [Liu, Shan-Lu](#)
To: [Saif, Linda](#); [Yount, Jacob](#)
Subject: Columbus Dispatch letter or commentary
Date: Saturday, March 21, 2020 4:03:36 PM
Attachments: [Dispatch commentary .docx](#)
[image001.png](#)

Hi Linda and Jacob:

Last few days, I have received numerous requests for interview, including local news media and even fire departments. I had to decline all of them for a variety of reasons. But I thought that it would be helpful for three of us to write a letter or commentary addressing some common questions and concerns people may have regarding the virus (not too much the COVID-19 disease). With this mind, I just had a draft and would share with you. I would appreciate your comments, edits, etc.

Again, this is just an idea and the draft is rough, kind of outline...

Thanks.

Shan-Lu



Shan-Lu Liu, M.D., Ph.D.
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Email: liu.6244@osu.edu; shan-lu.liu@osumc.edu

From: [Saif, Linda](#)
To: [Peter Daszak](#)
Subject: Re: Revised commentary for EMI - final!
Date: Tuesday, February 18, 2020 8:46:50 AM
Attachments: [image001.png](#)

Hi Peter,

Once again thanks for tackling this. Glad to see this group has reversed its conclusions once the actual data was analyzed and interpreted on a factual basis!

Hope this too gets to the "Grey House".

Thanks,

Linda

Linda J. Saif, PhD
Distinguished University Professor
Food Animal Health Research Program
OARDC/The Ohio State University
1680 Madison Ave
Wooster, Oh 44691

From: Peter Daszak <daszak@ecohealthalliance.org>
Date: Tuesday, February 18, 2020 at 12:27 AM
To: Linda Saif <saif.2@osu.edu>
Subject: RE: Revised commentary for EMI - final!

Definitely – already cited it. It's in review for Nature. Unfortunately this is the exact same group that elevated the potential that this was a lab release all the way to the White House two weeks ago, and helped fuel some of the conspiracy theorists.

'exact same group': P.O. authors: Andersen, Rambaut, etc.
P.O. was in review with Nature at the time.

Cheers,

Peter

Peter Daszak
President

EcoHealth Alliance
460 West 34th Street – 17th Floor

Cheers,

Peter

Peter Daszak

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EcoHealth Alliance leads cutting-edge research into the critical connections between human and wildlife health and delicate ecosystems. With this science we develop solutions that prevent pandemics and promote conservation.

From: Smriti Mallapaty [<mailto:smriti.mallapaty@nature.com>]

Sent: Friday, February 28, 2020 3:25 AM

To: Peter Daszak

Cc: Alison Andre; Aleksei Chmura; Robert Kessler

Subject: RE: nature news request

Dear Peter,

Just to follow up on this – do you know anyone who is seriously investigating this hypothesis? I would be interested in hearing more on this if any further research developments emerge.

Kind regards,

Smriti

From: Peter Daszak <daszak@ecohealthalliance.org>

Sent: Tuesday, 25 February 2020 4:26 AM

To: Smriti Mallapaty <smriti.mallapaty@nature.com>

Cc: Alison Andre <andre@ecohealthalliance.org>; Aleksei Chmura <chmura@ecohealthalliance.org>;
Robert Kessler <kessler@ecohealthalliance.org>

Subject: RE: nature news request

Hi Smriti,

The pig idea is based on:

- sequence analysis that shows the pig ACE2 receptor can likely bind with SARS-CoV-2, meaning it could likely infect pigs.
- Live pangolins are extremely rare in markets, so are unlikely to have played a significant role in transmission. Pangolin scales (dried, and therefore unlikely to be able to transmit virus) are normally sold.
- We still don't know the history of the pangolins that had the CoV with genetic elements close to SARS-CoV-2, and it's possible they were infected during transit from another intermediate host
- One plausible scenario is that there are farms with the virus circulating in a receptive mammal (e.g. a pig) in rural SW or Central China, and that these animals were taken to the wet markets, slaughtered and butchered, enhancing the transmission of the CoV into people.

Cheers,

Peter

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From: Smriti Mallapaty [<mailto:smriti.mallapaty@nature.com>]

Sent: Saturday, February 22, 2020 8:25 PM

To: Peter Daszak

Cc: Alison Andre; Aleksei Chmura; Robert Kessler

Subject: RE: nature news request

Dear Peter,

From: [Peter Daszak](#)
To: [Peter Daszak](#)
Cc: [Aleksei Chmura](#); [Robert Kessler](#); [Hongying Li](#)
Subject: Contact for journalists re. today's statement in the Lancet
Date: Tuesday, February 18, 2020 9:10:30 AM
Attachments: [20f11775Corr_2_Corrected_proofs.pdf](#)
Importance: High

Dear all,

We've been in touch with Lancet this morning to get all the final details of our statement of support letter fixed (see corrected proofs attached). We expect this to be published in about 1 hour from now, and have begun reaching out to journalists who might cover the story.

To make sure we have good coverage and a range of voices speaking with reporters, we would very much like to pass on your name and email address as someone for journalists to contact about the issues we're discussing in the letter – i.e. the need to support colleagues working under difficult situations in an outbreak, to reduce rumors and misinformation, and the scientific evidence demonstrating that the conspiracy theories on the virus' origins are unfounded.

If you are willing to talk with reporters, please send your phone numbers (landline and mobile if possible) by return to me and cc'ing Robert Kessler, who will help coordinate press interest. **Lancet will also be contacting journalists and we will pass on your contact info to them also.**

Ideally, we would have people from Europe, Australia, Asia, and the Americas (all represented on the authorship list) able to talk or email with journalists, so please step forward!

Cheers,

Peter

Peter Daszak

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From: [Peter Daszak](#)
To: [Peter Daszak](#)
Cc: [Aleksei Chmura](#); [Hongying Li](#)
Subject: URGENT - need signatures in next few hours: Our statement on COVID-19 will be published this morning US Eastern time in The Lancet
Date: Tuesday, February 18, 2020 5:51:48 AM
Attachments: [Statement of support COVID-19 China submission corrected clean.docx](#)
[TL_AuthorSigs 2019.pdf](#)
[icmje-COI-form to sign.pdf](#)
Importance: High

Dear All,

I want to let you all know that we received strong support from Richard Horton at *The Lancet*, and our paper will be published today (Tuesday 18th Feb) at 3pm UK time (10am Eastern US time). Thank you also to those of you who sent last minute changes – I've incorporated them where possible (see final version attached). I've also cited a paper that was uploaded yesterday (<http://virological.org/t/the-proximal-origin-of-sars-cov-2/398>), currently in review in *Nature* (I believe) that clearly refutes the bio-engineered virus hypothesis and strongly supports the conclusion that SARS-CoV-2 is of natural origin.

As we discussed, the authorship will be alphabetical. Unfortunately, it looks like there has to be a single corresponding author, but the editor will put a statement at the top of the authorship list to indicate that we are all speaking in one voice on this. I will see what that looks like when proofs come through in a minute. *The Lancet* have also agreed to publish our Mandarin version of this statement (thanks for the translation Hongying) online, so it reaches a wider audience in Asia and around the world.

I have two urgent requests:

- 1) Please fill in the attached Conflict of Interest form ASAP
- 2) Please e-sign the Author signature form ASAP

It will be really important to get this message out to journalists once it's published. Finally, I would ask all of you who can post this to your websites, or on social media, or email to your colleagues, please do so.

Cheers,

Peter

Peter Daszak
President

EcoHealth Alliance
460 West 34th Street – 17th Floor

From: [Saif, Linda](#)
To: [Peter Daszak](#)
Cc: [Aleksei Chmura](#); [Hongying Li](#)
Subject: Re: COVID-19 statement of support for Scientists and Public Health Professionals in China, and a condemnation of conspiracy theories on the origin of the virus
Date: Sunday, February 16, 2020 11:11:24 PM
Attachments: [Liu et al EMI Commentary Revision Final-sls.docx](#)
Importance: High

Hi Peter,

Thanks again for taking the lead on this—I saw the segment about Cotton which makes this statement all the more timely.

Attached is a commentary from me, Susan Weiss and 2 of my US Chinese American colleagues that we just submitted to EMI.

Hopefully NAS will put together a task force to address COVID-19 and these issues, especially since NAS was active in sending NAS members (I went) and others to Wuhan and Harbin to tour the new BSL4 facilities and foster relations with our Chinese CAS counterparts.

Regards,

Linda

Regards,

Linda

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Wooster, Oh 44691

From: Peter Daszak <daszak@ecohealthalliance.org>
Date: Sunday, February 16, 2020 10:48 PM
To: Peter Daszak <daszak@ecohealthalliance.org>
Cc: Aleksei Chmura <chmura@ecohealthalliance.org>, Hongying Li <li@ecohealthalliance.org>
Subject: COVID-19 statement of support for Scientists and Public Health Professionals in China, and a condemnation of conspiracy theories on the origin of the virus

Dear All,

Firstly, I want to thank each of you for your kindness, academic integrity, and openness in signing this statement of support. I've attached the current version of it for your records, and you can see the list of the eminent public health scientists from 8 countries who have co-signed with you. These include former heads of government agencies, current heads of world class research groups and major organizations, members of the US National Academies of Medicine and Science, an Academician from Malaysia, an Officer of the Order of Australia, and others who have achieved great success in public health and infectious disease research.

Secondly, I want to mention that the situation as regards conspiracy theories has worsened over the last few days, having been given credence through reporting yesterday in a UK tabloid (*Express*), regurgitation on prime time TV yesterday in the USA by US senator **Tom Cotton**, and **discussions at a very high level within government** in the USA and China. At the same time, some colleagues in China have received violent threats to their families and themselves.

For these reason, and as a way to get our statement across directly to the senior leaders in the governments of China and around the world, Jeremy Farrar (Director of the Wellcome Trust, and co-signatory) suggested that I submit this letter to the editor of *The Lancet* for possible publication. I have done so just now, out of a sense of urgency, and await his response (I have also asked if he is willing to sign).

I realize that all of you agreed to having this letter published and distributed, and I believe this would be an extremely appropriate platform to do so. Please let me know if you feel otherwise by responding to this email. Note that, for equity and impact, I have assigned authorship of the statement alphabetically, and will ask that no one person act as corresponding author.

I will, of course, let you know of the response as soon as I hear back.

Cheers,

Peter

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happens next week, it might get some pick up in the news. Is it OK for me to suggest your names to journalists as people who are knowledgeable on COVID-19, CoVs, pandemics, high impact viral diseases etc.? I think this paper helps a little bit to show that there's nothing unusual about COVID-19 being a bat virus that got into people naturally. There's also the fact that the other close virus (RmYN02) is in the same clade as SARS-2 and RaTG13, and that virus has an insertion in the Spike protein, disproving one of the conspiracy theories, but that won't stop them.

I feel like I should add a disclaimer to emails along the lines of "if you do speak to journalists about this work you may end up being targeted by nutjobs and potentially have one or two of your grants terminated by El Guru-in-Chief!"

BTW – please use Ralph's gmail address so I don't get FoIA'd

Cheers,

Peter

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EcoHealth Alliance develops science-based solutions to prevent pandemics and promote conservation

Talking points from Latinne et al. Origin and cross transmission of bat CoVs in China

In press at *Nature Communications*. Already peer-reviewed, this version is almost final. Uploaded to bioRxiv June 1st

Natural origin of COVID-19. The paper provides further evidence that SARS-CoV-2 emerged naturally from bats, casting even more doubt on the lab-origin conspiracy theory:

- The high diversity of bat-CoVs in the wild (781 novel genetic sequences) with 2 known to infect people, others known to infect human cells, and one that has caused an outbreak in pigs suggests ample opportunity for spillover of bat-CoVs into people on a regular basis.
- We find particularly high diversity of the group that contains SARS-CoV & SARS-CoV-2 (106 novel genetic sequences from *Sarbecovirus* genus – SARS-related CoVs).
- Pangolin viruses are close to SARS-CoV-2, but two bat viruses are more closely related, supporting its origin in bats (RaTG13 & RmYN02 – both from *Rhinolophus* horseshoe bats in S. China).
- SARS-CoV-2 has a furin-cleavage insertion between the S1 and S2 genes of the spike protein (the part that binds to human cells) which some have suggested is bioengineered. RmYN02 from bats has a similar insertion and is in the same clade as RaTG13 and the pangolin CoVs, proving that these insertions occur naturally within the same clade of viruses, even though RaTG13 doesn't have this.

Geographic origin of COVID-19. SARS-CoV-2 (cause of COVID-19) likely emerged from a clade of viruses in bats in SW China (Yunnan province) or countries bordering Yunnan (Myanmar, Lao PDR, Vietnam)

- This is the most thorough analysis of viruses related to SARS and COVID-19 ever published. We analyzed 781 novel genetic sequences that we discovered, published here for the first time, along with 509 previously known bat-CoVs. This means our conclusions on origin are the most accurate.
- However, we only sampled bats within China and the bats carrying closest relatives of SARS-CoV-2 (*Rhinolophus affinis* & *R. malayanus*) also occur in neighboring countries, across SE Asia and into S. Asia. It is likely that the same or similar viruses occur in these bats in other countries.
- Many of sampling sites for SARS-CoV positives were close to the border of Myanmar and Lao PDR.
- SW China was Quaternary glacial refugium for bat species incl. *Rhinolophus* spp. & may have allowed survival of older viral strains leading to increased diversity.

Importance for pandemic risk.

- This work was funded by the NIH grant that was recently terminated 'for convenience' following political interference from the White House - as reported in 60 Minutes and other outlets, and recently criticized by 77 Nobel Laureates and 31 Scientific Societies. All of the genetic sequences reported here are fragments, and our future plans were to sequence whole genomes and particularly the Receptor Binding Domains to see if any of these viruses are likely able to infect humans. That work will not happen without the funding from NIH.
- There is an high diversity of bat coronaviruses in southern China, some of which have already emerged in people and livestock, others that are poised to, still others about which we know very little. This represents a significant potential pandemic risk, and threat to food security through livestock disease. Even though we have found a few hundred new CoVs, we expect there to be many more across SE Asia and globally (perhaps as many as 10-15,000 bat-CoVs yet-to-be-discovered).
- Bats in SE Asia should be targeted for focused surveillance/viral discovery to help identify novel coronaviruses that may emerge in future. Sequences can be used to test vaccines, drugs. Control programs to stop them emerging can be targeted to where the risk is highest.